221-2327 Assignment #01:-

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Physical model & Symbols

$$V(s) = (Ls + R)I(s) + K_e \Omega(s)$$

(Electrical equation (armature circuit))

Mechanical eq. (rotor dynamics)

$$(Js + b)\Omega(s) = K_t I(s)$$

Where

$V(t)$ → input armature voltage. $V(s)$ its laplace transform
$I(s)$ → armature current
$\Omega(s)$ → angular velocity (rad/s)
$\Theta(s)$ → $\Omega(s)/s$ → angular position (rad)
R → armature resistance (Ω)
L → armature inductance (H)
J → rotor inertia (kgm^2)
b → viscous friction/damping (N.m.s)
K_t → torque constant (N.m/A)
K_e → back emf constant (V/(rad/s))

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Derive transfer function $\frac{\Omega(s)}{V(s)}$

$$I(s) = \frac{(J_s + b)}{k_t} \Omega(s)$$

$$V(s) = (Ls+R)I(s) + K_e \Omega(s) = (Ls+R)\left(\frac{J_s+b}{k_t}\right)\Omega(s) + k_e \Omega(s)$$

Factor $\Omega(s)$

$$V(s) = \left[\frac{(Ls+R)(J_s+b)}{K_t} + K_e\right]\Omega(s)$$

$$\frac{\Omega(s)}{V(s)} = \frac{k_t}{(Ls+R)(J_s+b) + K_t K_e}$$

∵ Second order denominator in s.

denominator is quadratic in s.

Transfer function $\frac{\Theta(s)}{V(s)}$ (position)

$$\frac{\Theta(s)}{V(s)} = \frac{\Omega(s)}{s}$$

$$= \frac{k_t}{s(Ls+R)(J_s+b) + K_t K_e}$$

one order higher (adds a pole at origin) — position

TF has an extra 1/s factor.

Neglect Inductance L (armature Inductance small)

Set $L=0$

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{R(Js+b)+K_tK_e} = \frac{K_t}{JRs+(bR+K_tK_e)}$$

first order system in s for velocity

Pole at $\frac{-bR+K_tK_e}{JR}$

Position TF becomes

$$\frac{\Theta(s)}{V(s)} = \frac{K_t}{s(JRs+(bR+K_tK_e))}$$

Which is second order system.
(one integrator + one real pole)

∴ if all physical parameters are positive the poles will have negative real parts.

∴ If poles are real positive, system is unstable.